1 Introduction

The inventory routing problem (IRP) arises in vendor-managed inventory systems, in which a supplier monitors the stock levels of its customers and takes responsibility for replenishing them. Over a multi-period planning horizon, the supplier jointly decides when each customer is visited, how much product is delivered at each visit, and how the vehicles are routed, subject to customer storage capacities, vehicle capacities, and the requirement that no customer runs out of stock. Integrating these decisions, rather than fixing replenishment plans first and routing them afterwards, can produce substantial savings (Archetti and Speranza, 2016). However, the integration couples a vehicle routing problem in every period with inventory dynamics that link the periods together. As a result, the IRP remains one of the most challenging problems in distribution logistics (Moin and Salhi, 2007; Bertazzi and Speranza, 2012; Coelho et al., 2014; Ekşi et al., 2026).

Most of the IRP literature minimises the total cost, defined as the sum of routing and inventory holding costs. In many applications, however, holding costs are difficult to estimate or are negligible compared with transport costs, and practitioners evaluate distribution plans by the transport cost per unit of product delivered. This key performance indicator, known as the *logistic ratio*, has long been used in the distribution of industrial gases (Bell et al., 1983; Benoist et al., 2011; Singh et al., 2015). It was also the objective of the ROADEF/EURO Challenge 2016, proposed by Air Liquide (André et al., 2020). Archetti et al. (2017b) formalised the IRP with logistic ratio (IRP-LR) and showed that solutions that minimise total cost can be poor in terms of the logistic ratio. One reason is the finite-horizon effect: under a cost objective there is little incentive to deliver product near the end of the horizon if it will only be consumed afterwards. The logistic ratio, in contrast, rewards every unit delivered and therefore produces replenishment plans that differ markedly from cost-minimising ones.

The IRP-LR is harder to solve than the classical IRP. Its objective is the ratio of two decision-dependent quantities: the routing cost in the numerator and the total delivered quantity in the denominator. A move that shortens the routes can therefore worsen the objective if it also reduces the quantity delivered, and a longer route can be preferable if it allows a larger delivery. Exact approaches build on parametric fractional programming (Dinkelbach, 1967), which replaces the ratio with a sequence of linear objectives. Archetti et al. (2017b) and Archetti et al. (2019) developed exact algorithms for the IRP-LR along these lines, and Baldacci et al. (2020) proposed a general exact framework for vehicle routing problems with fractional objectives. These methods provide optimal solutions and bounds, but their computational requirements restrict them to instances with at most 50 customers. For larger instances, to the best of our knowledge, the only heuristics evaluated on the IRP-LR benchmark are the iterated local search (ILS) and simulated annealing (SA) algorithms of Alvarez et al. (2018). The algorithms developed for the ROADEF/EURO Challenge 2016 address a richer industrial variant with a ratio objective, using heuristic

branch-cut-and-price (Absi et al., 2020), local search combined with mathematical programming (Su et al., 2020), fixed-sequence reoptimisation of quantities and timing (He et al., 2020), and hyper-heuristic sequence selection (Kheiri, 2020). More recently, ratio objectives have been studied in a multi-source setting solved by ILS (Alonso-Pecina et al., 2024) and in a multi-product setting with lateral transshipment and uncertain demand (Man et al., 2025). Compared with the classical IRP, for which a wide range of exact and heuristic methods exists, the heuristic literature on the IRP-LR remains sparse.

1.1 Related literature and research gaps

For the classical IRP, exact methods based on branch-and-cut (Archetti et al., 2007; Coelho and Laporte, 2013), branch-price-and-cut (Desaulniers et al., 2016), and their matheuristic extensions (Skålnes et al., 2023) solve instances of moderate size. Large instances are handled by heuristics, which are reviewed by Sofianopoulou and Mitsopoulos (2021) and Ekşi et al. (2026). We organise the discussion around three design choices that are particularly important under a logistic-ratio objective. Table 1 summarises the studies most closely related to this work.

Where delivery quantities are decided. Efficient local search is available for routing decisions, whereas delivery quantities are naturally determined by mathematical programming. Most IRP heuristics therefore decompose the problem: a heuristic explores visit schedules and routes, and a mathematical program sets or repairs the quantities (Iswari et al., 2026). Examples include the tabu search combined with mixed-integer programming of Archetti et al. (2012), the matheuristic of Archetti et al. (2017a), the variable MIP neighbourhood descent of Coelho et al. (2020), the ILS matheuristic of Laganà et al. (2024), and route-improving matheuristics for related production routing problems (Vadseth et al., 2023). In the hybrid genetic search (HGS) of Castro et al. (2026), the routing component builds on the HGS for the capacitated vehicle routing problem (CVRP) (Vidal, 2022), while the quantities are obtained by solving a minimum-cost flow problem with the network simplex algorithm. Zhao et al. (2025) extend HGS with a large neighbourhood that simultaneously changes visit periods, insertion positions, and delivery quantities across the horizon. Decomposition works well when holding costs make quantity decisions secondary to routing. Under the logistic ratio, however, the change in delivered quantity caused by a routing move is part of the value of that move. A move that is evaluated with quantities held fixed can appear unattractive even though it becomes improving once the quantities are adjusted, and solving a mathematical program for every candidate move is too expensive within a local search. This suggests evaluating each move by its combined effect on routing cost and delivered quantity. It also suggests using an exact optimisation step for quantities only occasionally, as He et al. (2020) showed with fixed-sequence reoptimisation.

How infeasibility is handled. Allowing a search to visit infeasible solutions and penalising the violation in the objective is a well-established strategy in vehicle routing. It is a central component of HGS (Vidal, 2022), and in inventory routing it has been used to relax vehicle capacity (Archetti et al., 2012; Castro et al., 2026) and customer stockouts (Zhao et al., 2025). In the IRP, the stockout constraints make the feasible region particularly fragmented. Removing a single visit usually causes a stockout in some later period, so a search restricted to feasible solutions cannot easily move between replenishment plans with different visit patterns. Penalising stockouts addresses this, but under a ratio objective the penalty weight is difficult to set. The logistic ratio is measured in cost per unit and varies widely across instances, so a fixed penalty is either too weak, and the search remains infeasible, or too strong, and the search behaves as if infeasibility were forbidden. Lagrangian relaxation (Fisher, 1981) offers a principled alternative: the stockout constraint is dualised, and the multiplier is updated with a subgradient-type rule that scales the penalty to the current gap from the incumbent. To the best of our knowledge, this has not previously been used to guide a local search for the IRP-LR.

Saturated vehicles. Because the logistic ratio rewards large deliveries, vehicles in good IRP-LR solutions are typically fully loaded. Any move that inserts a customer into such a vehicle is then infeasible if the existing quantities are kept, so large parts of the inter-route and inter-period neighbourhoods are effectively unavailable. In the CVRP, HGS addresses the corresponding issue by allowing capacity violations at a penalty (Vidal, 2022). In the IRP-LR, there is a feasible alternative. The deliveries already on a full vehicle often include quantities that are not needed to prevent future stockouts, and this slack can be reassigned to the inserted customer without violating any constraint. Exploiting this slack keeps the search within capacity while giving access to the moves that matter most for the logistic ratio.

Together, these observations identify the gap addressed in this paper: there is no scalable heuristic for the IRP-LR that (i) evaluates routing and quantity decisions jointly within each move, (ii) explores inventory-infeasible solutions in a controlled manner that adapts to the ratio objective, and (iii) does not depend on a mathematical programming solver.

1.2 Contributions

This paper proposes a multi-start iterated local search (MSILS) for the IRP-LR. The algorithm does not require a mathematical programming solver. Its main contributions are as follows.

1. **Joint evaluation of route and quantity decisions.** Four of the five local search operators update delivery quantities as part of each move, using closed-form rules that respect depot supply, vehicle capacity, and customer storage capacity. Every candidate move is therefore evaluated by

Table 1: Studies most relevant to the IRP-LR and to the proposed solution methodology.

Study	**Problem / objective**	**Solution approach**	**Key search characteristic**	**Relevance to this work**
Archetti et al. (2017b)	IRP; routing cost / quantity delivered	Exact fractional optimisation	Joint routing and replenishment	Introduces the IRP-LR
Alvarez et al. (2018)	IRP including logistic ratio	ILS and SA	Neighbourhood search over routes and replenishment	Benchmark heuristic for the IRP-LR
Archetti et al. (2019)	IRP-LR	Iterative exact method	Sequence of IRPs with linear objectives	Optimal solutions for small instances
Baldacci et al. (2020)	VRP with fractional objective	Exact method	Primal and dual bounding procedures	General fractional routing methodology
Absi et al. (2020)	Rich IRP; unit distribution cost	Heuristic branch-cut-and-price	Integrated routing, quantity, and scheduling	Large-scale industrial ratio IRP
Su et al. (2020)	ROADEF/EURO IRP	Local search with MIP/LP	Mathematical programming sets timing and quantities	Hybrid local-search design
He et al. (2020)	ROADEF/EURO IRP	Fixed-sequence matheuristic	Reoptimises quantities and timing for fixed visit sequences	Separation of route structure and quantity optimisation
Alonso-Pecina et al. (2024)	Operating cost / quantity delivered	ILS	Neighbourhood search for a multi-source IRP	Recent ILS with a ratio objective
Man et al. (2025)	Multi-product logistics-ratio IRP	Robust fractional optimisation	Uncertainty and lateral transshipment	Extension of logistics-ratio optimisation
Castro et al. (2026)	Classical IRP	HGS	Route search with min-cost-flow quantity optimisation	State-of-the-art decomposition-based search
Zhao et al. (2025)	Classical and stockout-penalised IRP	HGS with large neighbourhood	Cross-period route/quantity moves; penalised stockouts	Joint moves and controlled infeasibility
This paper	IRP-LR	Multi-start ILS (solver-free)	Joint route–quantity moves; vehicle slack; Lagrangian stockout penalty	Scalable heuristic for the IRP-LR

its actual effect on the logistic ratio, that is, on routing cost and delivered quantity together.

2. **Vehicle slack.** We introduce a mechanism that reassigns quantity not needed to prevent future stockouts from the existing deliveries of a fully loaded vehicle to a newly inserted customer. The mechanism is motivated by the capacity-violation strategy of HGS (Vidal, 2022), but it never violates vehicle capacity. It makes available the inter-route and inter-period moves that would otherwise be rejected, and it creates no new stockouts.

3. **Lagrangian-guided exploration of infeasible solutions.** The zero-stockout constraint is dualised, and the local search solves a sequence of relaxed problems in which the Lagrange multiplier is updated by a subgradient-type rule based on the gap from the incumbent. The penalty therefore scales automatically to the magnitude of the logistic ratio of each instance, allowing the search to cross infeasible regions and then return to feasibility.

4. **Problem-specific construction, rebalancing, and perturbation.** Initial solutions are built by a randomised constructive heuristic that prioritises customers by stockout urgency and distance from the depot. A rebalance procedure redistributes quantities over a fixed schedule whenever the incumbent improves, and an adaptive perturbation increases its strength as the number of non-improving iterations grows.

5. **Computational evidence.** Experiments on 1128 benchmark instances, comprising 978 instances with 5 to 50 customers and 150 instances with 50 to 200 customers, show that the MSILS improves on the ILS and SA of Alvarez et al. (2018) by on average 6.30% and 8.00%, respectively, on the large instances, with larger gains as the instance size grows. On small instances, it finds solutions within X% of the optimal values reported by Archetti et al. (2019). A component analysis quantifies the contribution of each mechanism, and we also examine whether embedding the HGS of Vidal (2022) as a routing post-optimiser brings further improvement. The instances are publicly available (and the code will be released upon acceptance).

The remainder of the paper is organised as follows. Section **??** defines the IRP-LR and the Lagrangian-relaxed problem used by the algorithm. Section **??** describes the MSILS. Section **??** reports the computational experiments, and Section **??** concludes and outlines directions for future research.

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